MILITARY SPECIFICATION

THERMOMETER, SELF INDICATING, LIQUID IN GLASS, INCUBATOR, 0^{0} to 70^{0} C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of partial immersion mercury in glass thermometer suitable for use in medical incubators (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

NBS Monograph 150 - Liquid-in-Glass Thermometry.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Material. Material shall be of high quality throughout and suitable in all respect for the purpose intended.

3.1.1 Glass tubing. Glass tubing shall have a diameter of between 6.0 to 7.0 millimeters (mm) (see 3.2.1).

3.1.1.1 Bulb. Glass for bulb tubing shall be as specified in NBS Monograph 150.

3.1.1.2 Stem. Glass for stem tubing shall have physical properties which are compatible with the glass in the bulb.

3.1.2 Mercury. The mercury shall be clean, dry, and free of foreign material and entrapped gas.

3.1.3 Gas. The gas used to fill the space above the mercury shall be dry, oxygen free, and shall not react with the mercury.

3.2 Construction. The thermometer shall be mercury column in glass type, enamel-backed, graduated from 0° to 70° Centigrade (C) in $1/2^{\circ}$C subdivisions. There shall be an integral glass collar with a diameter that is at least 1.8 mm larger than the diameter of stem.

3.2.1 Stem and bulb. The diameter of the bulb shall be the same or less than the diameter of the stem. The diameter of the bore shall be no less than 0.10 mm. Diameter shall be uniform throughout the graduated length and shall be as specified in Table I. An expansion chamber shall be located at the top of the bore which shall permit the thermometer to reach a temperature of at least 100°C without breaking when the thermometers are tested as specified in 4.4.1. The bore shall be clean and moisture free.

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3.2.2 Graduations and numerals. The thermometer shall be graduated in the Celsius scale. The graduation lines shall be clear cut, straight, narrow, of uniform width and shall be perpendicular to the longitudinal axis of the thermometer. Width of graduation lines shall not exceed 0.2mm. The distance between centers of graduation lines shall be no less than 0.4mm. Major graduations shall be spaced and numbered at 10° intervals. Major graduations shall be longer than minor graduations. Graduation lines, numerals and identification marks shall be black except 37° and 56° graduations, which shall be red or shall be indicated by an arrow. The markings shall show no change when tested as specified in 4.4.3. The smallest interval shall be 0.5 degree Celsius.

3.2.2.1 Position of end graduations. The bottom graduation mark shall be placed so there is not less than 13mm of uniform bore below the mark. The top graduation shall be placed so that there is at least 10mm of uniform bore above the mark.

3.3 Dimensions and tolerances. The thermometer shall have dimensions and tolerances as shown in table I. Any dimension not within the tolerances specified herein shall be classified as a defect.

TABLE I. Dimensions and tolerances.

	Dimensions (mm)	Tolerances (mm)
Stem diameter	6.5	±0.5
Overall length	335	± 5
Distance from bottom of thermometer to bottom of collar	174	± 4
Distance from bottom of collar to bottom of scale	25	+10 − 5
Scale length	100	minimum
Distance from top of scale to bottom of expansion chamber	10	minimum

3.4 Calibration. Each thermometer shall be calibrated at a stem immersion of 170mm ± 2mm to an accuracy of ± 0.5^{o}C.

3.5 Accuracy. Thermometer shall not vary more than 0.5^{o}C from the applicable graduation, when tested as specified in 4.4.2.

3.6 Strain. When tested as specified in 4.4.4, the bulb and adjacent areas of the stem shall be free from ring and longitudinal strain, and any strain in the remainder of the stem shall not exceed Temper No. I. Diffuse strain in the upper portion of the stem, when examined under polariscope, shall not be cause for rejection. However, the strain in the upper portion of the thermometer shall be adequately diffused so that the strain will not have any detrimental effect on the thermometer's accuracy or cause the thermometer to crack.

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3.7 Workmanship. Workmanship shall be first class throughout. Thermometers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, III, IV, (see 4.2.1, 4.2.2 and 4.3) and 4.3.1.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table II. Unit of product for examination purposes shall be one thermometer.

TABLE II. Sampling for examination.

Examination	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5
For dimensional examination	S-1	*

* Acceptance number shall be zero.

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Thermometers shall be tested according to table III. Unit of product for test purposes shall be one thermometer.

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TABLE III. Tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Heat tolerance	3.2.1	4.4.1	S-2	*
Accuracy	3.5	4.4.2	S-3	*
Permanency of marking	3.2.2	4.4.3	S-1	*
Strain	3.6	4.4.4	S-3	*

* Acceptance number shall be zero.

4.2.2.1 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The thermometer shall be examined for defects including, but not limited to, those listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major	
101	Thermometers not free of sharp edges.
102	Thermometer not free of cracks.
103	Thermometer not free of chips.
104	Graduations and graduation markings not complete, correct, or legible.
105	Glass not free of bubbles.
106	Mercury column not free of breaks.
Minor	
201	Glass not free of stones, seeds, and strae.
202	Thermometer not free of scratches.

4.3.1 Packaging inspection. The inspection of unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with test characteristics.

4.4.1 Heat tolerance. The thermometers shall be immersed for a distance of 170mm ± 2mm in a bath for 5 minutes at 100°C.

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4.4.2 Accuracy. Each thermometer shall be tested in a suitable testing bath at 20°C, 40°C and 60°C against a standard thermometer, the accuracy of which shall be traceable to NBS certification.

4.4.3 Permanency of marking. The marking shall be exposed to 20 percent caustic soda at 65°C + 2°C for 1/2 hour, followed by exposure to 50 percent sulfuric acid at 65°C ± 2°C for 1/2 hour. This cycle shall be repeated twice.

4.4.4 Strain. The strain on the thermometer shall be determined in accordance with ASTM C148, Method A or B.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each thermometer shall be packaged in a spiral-wound paper tube, plastic box or tube or a box conforming to PPP-B-676, type I. The container shall contain sufficient cushioning material to insure against damage to the thermometers. Closure of the container shall be adequate to prevent accidental opening.

5.1.1.2 Intermediate package. Twelve unit packages shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Each thermometer shall be packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twenty-four intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-585, class 3, style 3; PPP-B-601, overseas type, PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Twenty-four intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged thermometers shall be packed in shipping containers in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers' rules and regulations.

5.3 Marking. Each unit package, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. This thermometer is intended for use with the following items:

National Stock No.	Item Identification
6640-00-428-8120	INCUBATOR, BACTERIOLOGICAL, Large, 115-230 volt, 60 Hz, AC
6640-00-428-9120	INCUBATOR, BACTERIOLOGICAL, Small, 115-230 volt, 60 Hz, AC

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. The thermometer covered by this specification is listed in the Federal Supply Catalog under the following stock number.

National Stock No.	Item Identification
6685-00-299-8496	THERMOMETER, SELF-INDICATING, LIQUID IN GLASS, Incubator, 0° to $70^{o}C$

6.4 Limitations. This specification does not cover all types of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6685-0579

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-37594 THERMOMETER, SELF INDICATING, LIQUID IN GLASS, INCUBATOR, 0^{0} to 70^{0}

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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